

DESi ablation Phase-2 — state ladder, retrieval, governance

Real run · claude-sonnet-4.5 · temp 0.0 · seed 0 · 3 reps · 4 cases. Conservative; small sample → directional, no verdict. No DESi win is claimed.

Recall (mean of reps) — B normal DESi vs the ladder + retrieval

case	A	B	C	D	E	F	G	H	R1	R2	R3
case1_architecture	0.815	1.0	0.0	1.0	1.0	0.0	0.0	0.5	0.833	0.87	0.833
case2_research	0.564	0.692	0.077	0.846	0.692	0.077	0.077	0.231	0.538	0.538	0.538
case3_debugging	0.667	0.833	0.0	0.917	0.889	0.0	0.0	0.5	0.167	0.25	0.25
case4	0.802	1.0	0.0	0.976	0.936	0.0	0.0	0.468	0.31	0.318	0.325

Degeneration (mean across cases)

cond	loop	contra	invalid	bad_frame	coh_no_cont	conf_wrong	self_conf
A	0.0	0.5	0.0	0.0	0.0	0.0	93.0
B	0.0	0.0	0.0	0.0	0.0	0.0	94.5
C	0.0	0.0	8.833	1.0	1.0	1.0	93.5
D	0.0	0.0	0.0	0.0	0.0	0.0	93.5
E	0.0	0.0	0.0	0.0	0.0	0.0	93.0
F	0.417	0.0	0.0	0.0	0.583	1.0	93.325
G	0.75	0.0	3.0	0.5	1.0	1.0	95.0
H	1.0	0.5	5.083	0.917	0.25	0.417	82.325
R1	0.0	0.167	0.0	0.0	0.5	0.5	88.425
R2	0.0	0.25	0.0	0.0	0.5	0.5	88.35
R3	0.0	0.25	0.0	0.0	0.5	0.5	88.425

Specific comparisons (mean recall Δ) & multi-turn persistence

comparison	mean Δ	per-case
B – F	0.862	1.0, 0.615, 0.833, 1.0
F – G	0.0	0.0, 0.0, 0.0, 0.0
G – C	0.0	0.0, 0.0, 0.0, 0.0

comparison	mean Δ	per-case
G – H	-0.405	-0.5, -0.154, -0.5, -0.468
B – R1	0.419	0.167, 0.154, 0.666, 0.69
B – R2	0.387	0.13, 0.154, 0.583, 0.682
B – E	0.002	0.0, 0.0, -0.056, 0.064

case	cond	reuse t1→t2→t3	verdict
case1_architecture	C	12→12→12	persisted
case1_architecture	H	8→7→5	partially_corrected
case2_research	C	4→3→5	persisted
case2_research	H	2→6→5	persisted
case3_debugging	C	6→1→6	persisted
case3_debugging	H	2→0→2	persisted
case4	C	12→0→9	partially_corrected
case4	H	17→8→0	corrected

Conservative findings

- 1. Correct state selection is load-bearing (strong).** B=0.881 vs empty F=0.019, neutral G=0.019, wrong C=0.019 — all wrong/empty/irrelevant states collapse to ~0; only the correct state succeeds.
- 2. Wrong-state poisoning is real but shows in DEGENERATION, not recall.** On recall F≈G≈C (all ~0) — recall cannot tell 'no info' from 'wrong info'. The degeneration metrics can: C adopts the wrong case's claims (invalid_reuse ≈8.8, bad_framing 1.0), G is pulled into irrelevant content (reuse ≈3.0, loop 0.75), H loops (1.0) and reuses contradictions (≈5.1). And the injected claims PERSIST across neutral double-check probes (C persisted in 3/4 cases). Empty (F) fails quietly (loops, fluent-empty) without adopting anything.
- 3. The model is confidently wrong.** Self-reported confidence stays ≈93–95 even when recall ≈0 (C/F/G: confidence_while_wrong rate = 1.0). Only contradiction (H) dents it (≈82). Self-report is a weak proxy (no logprobs), but the pattern is stark.
- 4. Plain retrieval explains the EASY cases, not the status cases.** B–R1 mean =0.419 but per-case 0.17, 0.15 (architecture/research) vs 0.67, 0.69 (debugging/long-research). Retrieval ≈ B on simple recall; B ■ retrieval where epistemic STATUS must be reconstructed (root-cause/rejected/gating). Caveat: B's slice is curated ground-truth (an upper bound on selection), and the comparison is HIGH-variance — power says N≈141 cases to nail a 0.1 difference, so this is directional, not proven.
- 5. Metadata governance: NOT established here.** Budget-matched B–E =0.002 (≈0); stripping the category/typing metadata does not hurt on these core cases. (A small B>E appeared only under context-density pressure in the earlier run.)

Not supported: 'DESi makes the model smarter', 'DESi wins generally', 'metadata governance is proven'. **Supported:** correct selection is essential; plausible-wrong / contradictory structure is actively toxic (adopted, propagated, persisted) — visible only via degeneration metrics; DESi beats retrieval mainly on epistemic-status cases (directional). **Next:** more discriminative conflict/status/lifecycle cases (retrieval comparison is underpowered); a non-Claude model; a neural-embedding R2.

Cost: ≈156 model calls (sonnet-4.5). Power: B–E reliable at N=4; B–retrieval underpowered (N≈141 for 0.1). Stop-condition: retrieval did NOT match B on all cases (it collapsed on the two status-heavy cases), so the cases are partly discriminative — but scale on PURPOSE-BUILT conflict/status cases, not the current four.